

Lesson 05: Exact First-Order Differential Equations

Professor Dave Explains Differential Equations Master Series · Lesson 5

Video: Exact First-Order Differential Equations Course: Applied Ordinary Differential Equations (ENGR 213)

1. Video Intuition & Pedagogical Framing

Professor Dave explains that an exact equation is a differential equation disguised as the total derivative of a topographic height map $\phi(x,y) = C$. Just as walking along a contour line on a mountain means your altitude change is zero ($d\phi = 0$), an exact differential equation represents movement along the level curves of a multivariable potential function. By testing mixed partial derivatives $\partial M/\partial y = \partial N/\partial x$, we verify exactness and reconstruct the hidden potential function.

2. Core Theoretical Framework & Rigorous Formulations

The Exact Differential Mechanism

An equation in differential form $M(x,y)dx + N(x,y)dy = 0$ is exact if and only if:

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

When exact, there exists a potential function $\phi(x,y)$ such that $\frac{\partial \phi}{\partial x} = M$ and $\frac{\partial \phi}{\partial y} = N$. The general solution is simply $\phi(x,y) = C$.

Non-Exact Integrating Multipliers:

- If $\frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) = f(x)$, then $\mu(x) = \exp \left(\int f(x) dx \right)$.
- If $\frac{1}{M} \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) = g(y)$, then $\mu(y) = \exp \left(\int g(y) dy \right)$.

3. Fully Solved Whiteboard Problems

Classroom Problem: Step-by-Step Exact ODE Reconstruction

Problem Statement: Verify that $(2xy + 3)dx + (x^2 - 1)dy = 0$ is exact, and solve with initial condition $y(1) = 4$.

Step 1: Test Clairaut Exactness Condition

$M = 2xy + 3 \implies \frac{\partial M}{\partial y} = 2x$. $N = x^2 - 1 \implies \frac{\partial N}{\partial x} = 2x$. Since $\partial_y M = \partial_x N = 2x$, the equation is strictly exact.

Step 2: Partially Integrate M with respect to x

$\phi(x,y) = \int (2xy + 3)dx + g(y) = x^2y + 3x + g(y)$.

Step 3: Differentiate with respect to y and Equate to N

$\frac{\partial \phi}{\partial y} = x^2 + g'(y) = N = x^2 - 1 \implies g'(y) = -1$.

Step 4: Integrate $g'(y)$ and Assemble General Solution

$g(y) = -y$. Therefore, $\phi(x,y) = x^2y + 3x - y = C$.

Step 5: Apply Initial Condition

$y(1) = 4 \implies (1)^2(4) + 3(1) - 4 = C \implies C = 3$. Explicit form: $y(x)(x^2 - 1) + 3x = 3 \implies y(x) = \frac{3-3x}{x^2-1} = \frac{-3(x-1)}{(x-1)(x+1)} = \frac{-3}{x+1}$ for $x \neq -1$.

⚠ Common Exam Pitfall & Trapdoor Avoidance

When integrating $M(x,y)$ with respect to x , the constant of integration is an arbitrary function of y , denoted $g(y)$, NOT a scalar constant $+C$!

4. Key Takeaways & Exam Strategy Formula Card

🌟 Core Takeaway & Exam Rule

Test mixed partials first. Integrate M over x , match derivative to N to find $g(y)$, and set $\phi(x,y) = C$.

- Video reference: [Exact First-Order Differential Equations](#)
- Always verify boundary conditions and check that candidate solutions satisfy the governing differential equation identically.